

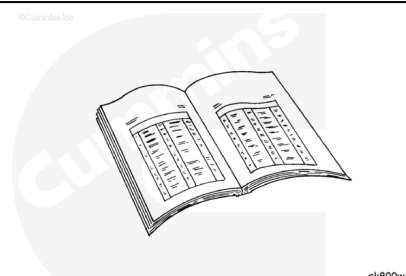
Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When performing any or all of the following procedures to remove fuel supply lines and related components, keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on a fuel system.



⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

- Remove as much of the engine wire harness and brackets as necessary to gain access to the fuel tubes. Refer to Procedures 019-043 (</qs3/pubsys2/xml/en/procedures/19/19-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System QSK19, QSK23, QSK45, QSK60 and QSK78 Engines, Bulletin 3666113 or 019-043 (</qs3/pubsys2/xml/en/procedures/13/13-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System Series QSK19 CM850 Modular Common Rail System Engine, Bulletin 4021493.
- On engines with an air compressor, remove the air inlet connection. Discard the connection hose and the gasket. Refer to Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html>).

with Electronically Actuated Injector

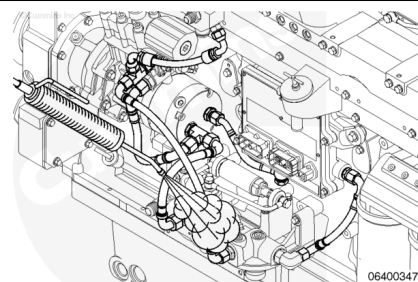
⚠ WARNING ⚠

Wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ CAUTION ⚠

Very small amounts of dirt and debris can be very harmful to the injectors and the cone seats on the injectors. Extra care is required to keep the fuel connections clean during removal and installation. The fuel supply line(s) must be placed in a plastic bag or connections be covered immediately to keep them clean when removed from the engine. Dirt and debris can cause fuel system damage.

- Thoroughly clean debris from all fittings and components before removal. Make sure that the debris, water steam, or cleaning solution does **not** get inside the fuel system.

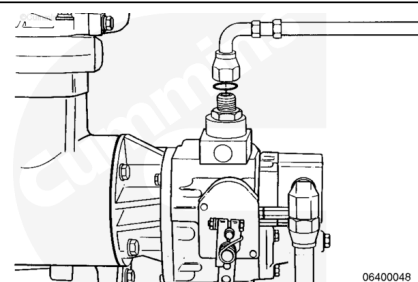


Remove

with Mechanically Actuated Injector

⚠ WARNING ⚠

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Use two wrenches to remove the fuel tubes and hoses.

Support the mating fittings with a wrench. Loosen the fuel tube nuts with the other wrench.

Loosen the fuel tubing nuts at the fuel junction block on the fuel manifold.

Remove the fuel tube clamps.

Remove the fuel tubing mounting clamp and the fuel tubes.

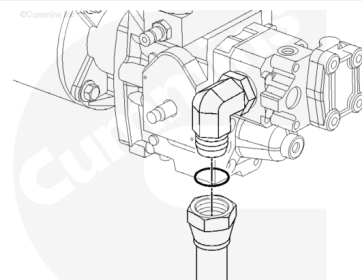
Remove the fuel control supply hose.

Discard the o-ring.

Remove the hose between the fuel filter head outlet and the fuel pump inlet fitting.

Discard the o-rings.

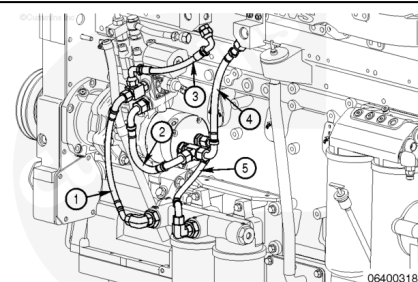
Remove the fuel inlet line from the fuel tank to the fuel filter.



with Electronically Actuated Injector

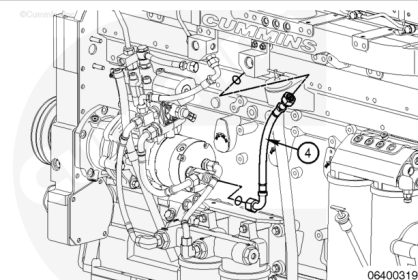
Fuel lines to be removed:

1. Stage 2 filter outlet to inlet manifold
2. Bypass from inlet actuator back to gerotor
3. Air bleed line
4. Fuel supply from fuel inlet manifold
5. Outlet of gerotor pump to stage 2 filter inlet.



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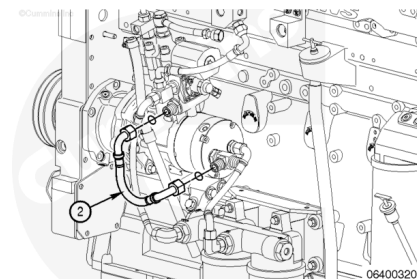


⚠ CAUTION ⚠

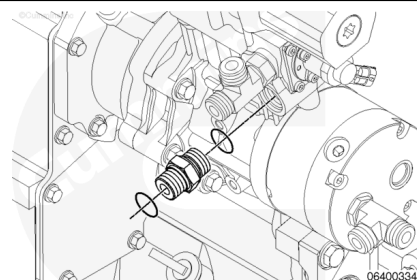
Very small amounts of dirt and debris can be very harmful to the injectors and the cone seats on the injectors. Extra care is required to keep the fuel connections clean during removal and installation. The fuel supply line(s) must be placed in a plastic bag or connections be covered immediately to keep them clean when removed from the engine. Dirt and debris can cause fuel system damage.

Disconnect the fuel supply line (4) from the elbow union on the fuel filter head assembly and from the tee union on the fuel pump. Discard the o-rings.

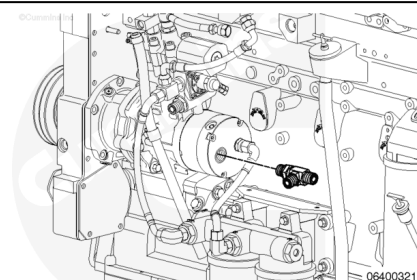
Disconnect the fuel supply line (2) from the fuel pump and from the tee fitting on the back of the fuel pump. Discard the o-rings.



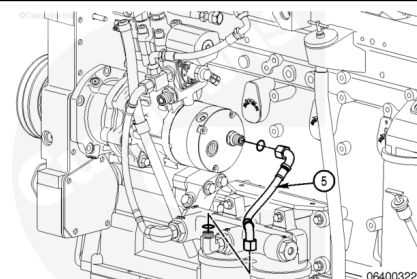
Remove the union from the fuel pump. Discard the o-rings.



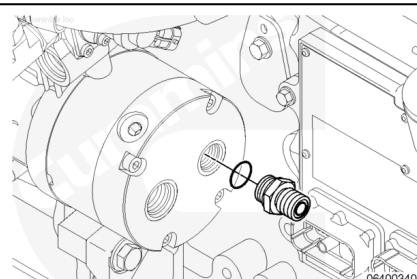
Remove the tee fitting from the back of the fuel pump. Discard the o-rings.



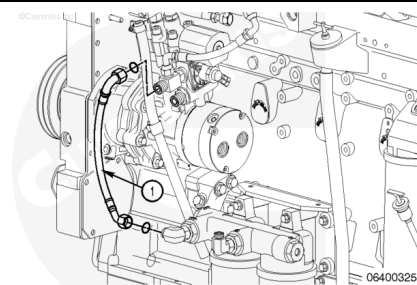
Remove the fuel supply line (5) from the back of the fuel pump and from the stage 2 fuel filter inlet.



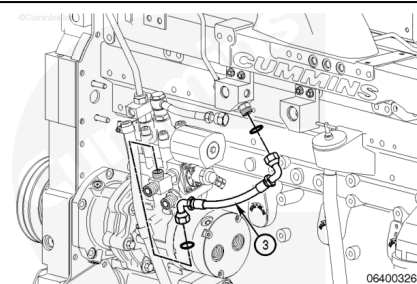
Remove the union from the back of the fuel pump. Discard the o-rings.



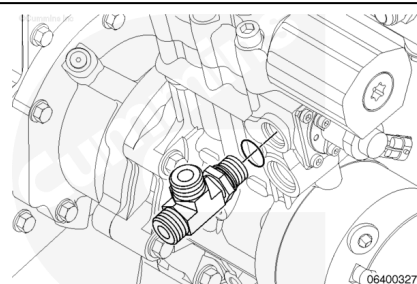
Disconnect the fuel supply line (1) from the fuel filter head union.



Disconnect the fuel supply line (3) from the tee fitting on the fuel pump and from the fuel return manifold. Discard the o-rings.



Remove the tee fitting from the fuel pump. Discard the o-rings.

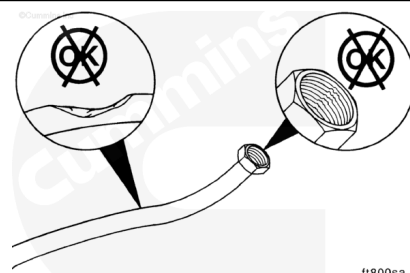


Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



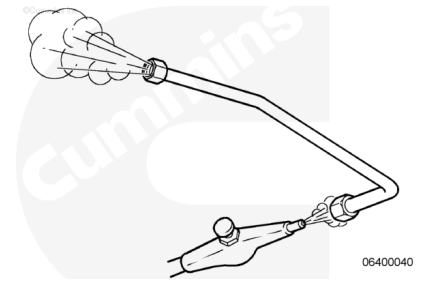
Remove and discard the ferrule seals.

Use solvent to flush the tubing and fuel block connectors.

⚠ WARNING ⚠

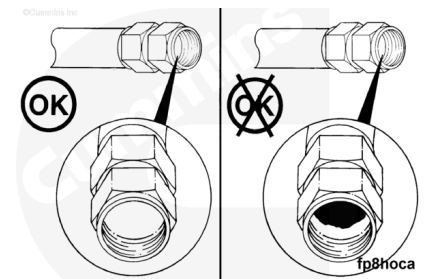
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use compressed air to dry the lines and flush any loose dirt particles from the lines.



Inspect the inside of the hose that connects the fuel tank to the fuel filter and the fuel inlet hose.

- The inner lining of the hose can separate from the center hose section.
- A separation or flap can cause a restriction in the fuel flow.

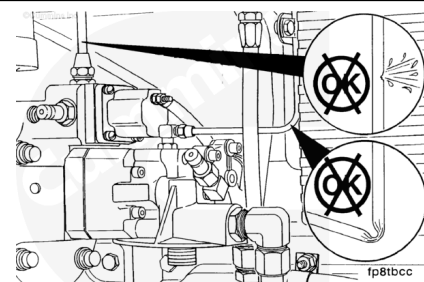


Inspect for any pinches in the hose that would obstruct the flow.

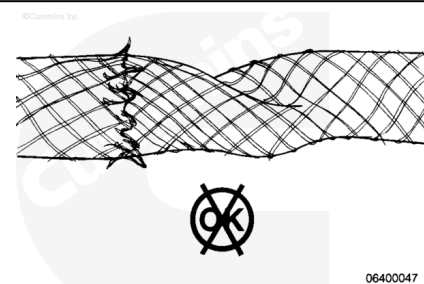
Make sure the hose does **not** have pinches or loops that would obstruct the flow.

Inspect the fuel line for cracks that can cause a loss of pressure.

Inspect the fuel line for sharp bends that can cause a restriction in pressure.



Inspect the fuel control supply hose for pinched areas that prevent the proper flow, or frayed mesh. The hose **must** be replaced, if damaged.



with Electronically Actuated Injector

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⚠ WARNING ⚠

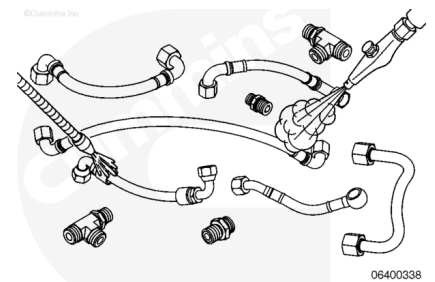
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

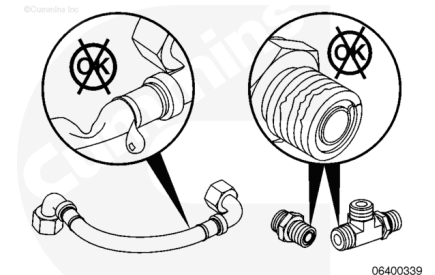
Use solvent to clean the fuel supply lines, tee fittings, and unions.

Dry with compressed air.



Inspect the fuel supply lines for restrictions that could possibly prevent fuel from flowing freely. Replace the fuel supply lines if kinks, damage, or leakage is found.

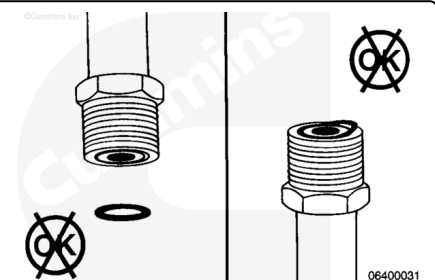
Inspect the tee fittings and unions for damage. Replace if damaged, cracked, or stripped threads are found.



Install

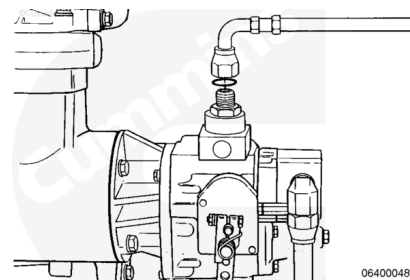
with Mechanically Actuated Injector

Install new o-rings on the control valve body inlet fitting and fuel pump outlet fitting. Make sure the o-rings are installed properly.

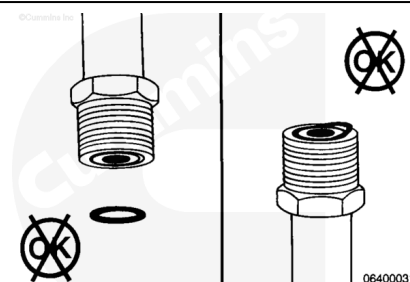


Install and tighten the fuel control valve supply hose.

Torque Value: 61 n•m [45 ft-lb]



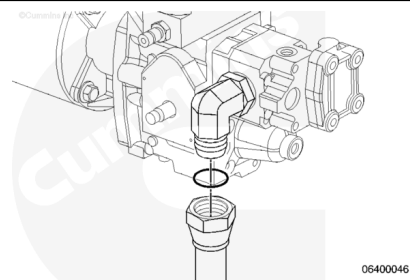
Install new o-rings on the fuel pump inlet line fitting and fuel filter head outlet fitting. Make sure the o-rings are installed properly.



Install the hose at the fuel pump inlet and fuel filter head outlet fitting.

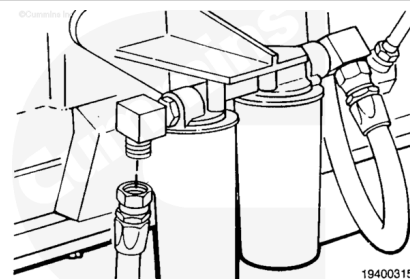
Tighten the hose.

Torque Value: 88 n•m [65 ft-lb]

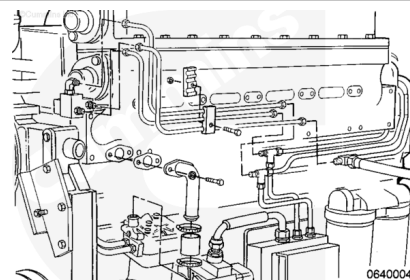


Install and tighten the fuel inlet line that connects the fuel tank to the fuel filter.

Torque Value: 115 n•m [85 ft-lb]



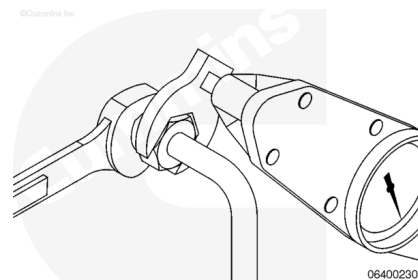
Loosely assemble the fuel tubes and clamps for the fuel rail and timing lines.



It is a good service practice to use two wrenches when tightening the fuel fittings.

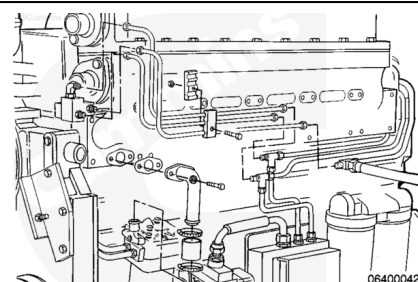
Support the fitting. Use a crowfoot wrench and a torque wrench to tighten the fuel tube nut.

Torque Value: 27 n•m [20 ft-lb]



Tighten the fuel tube clamps.

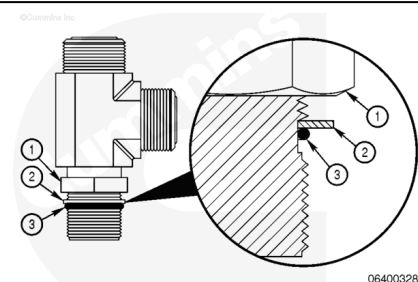
Torque Value: 27 n•m [20 ft-lb]



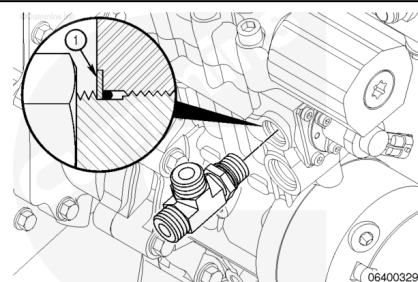
with Electronically Actuated Injector

Install the tee fitting in the top port of the fuel pump by first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.

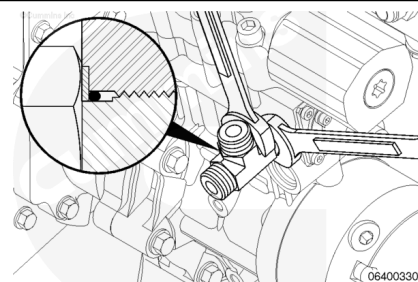


Install the tee fitting in the fuel pump port until the backup washer (1) contacts the port face.



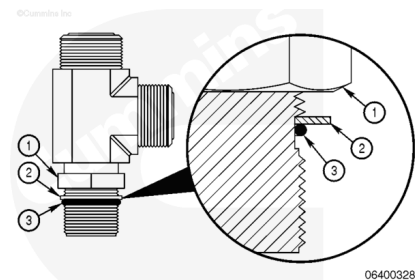
Using two wrenches, hold the tee fitting in the desired position and tighten the locknut.

Torque Value: 95 n•m [70 ft-lb]



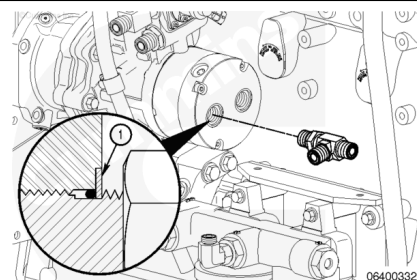
Install the tee fitting in the port on the back of the fuel pump by first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.



06400328

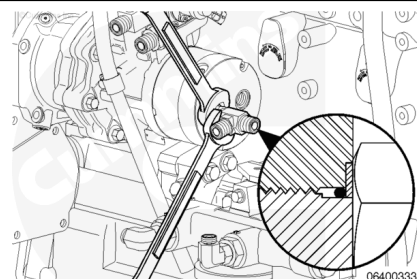
Install the tee fitting in the fuel pump port until the backup washer (1) contacts the port face.



06400332

Using two wrenches, hold the tee fitting in the desired position and tighten the locknut.

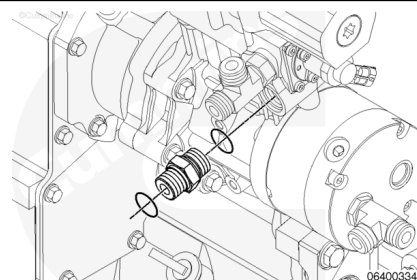
Torque Value: 95 n•m [70 ft-lb]



06400333

Install new o-rings on the union and install the union in the port next to the inlet actuator.

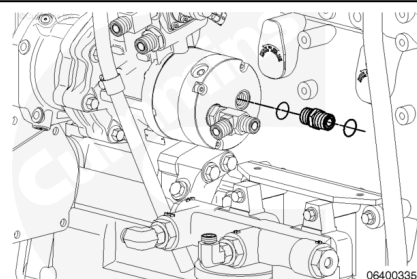
Torque Value: 95 n•m [70 ft-lb]



06400334

Install new o-rings on the union and install the union in the port on the back of the fuel pump.

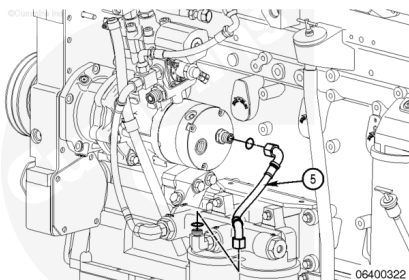
Torque Value: 72 n•m [53 ft-lb]



06400335

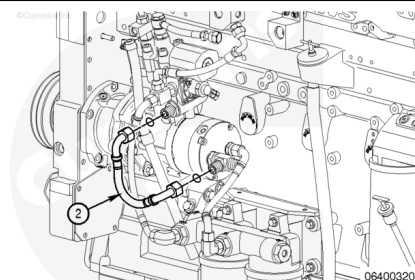
Install the fuel supply line (5) from the back of the fuel pump to the stage 2 fuel filter inlet.

Torque Value: 54 n•m [40 ft-lb]



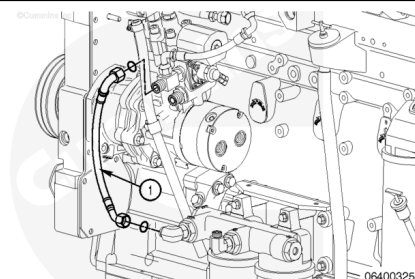
Install the fuel supply line (2) from the tee fitting on the back of the fuel pump to the union next to the fuel inlet actuator.

Torque Value: 65 n•m [48 ft-lb]



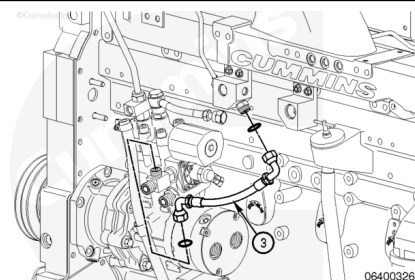
Install the fuel supply line (1) from the second stage fuel filter to the tee fitting next to the inlet actuator.

Torque Value: 45 n•m [33 ft-lb]



Install the fuel supply line (3) from the tee fitting next to the inlet actuator to the return manifold.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

with Mechanically Actuated Injector

- On engines equipped with an air compressor, install the air compressor air inlet connection with a new hose and gasket. Refer to Procedure 012-014 (</qs3/pubsys2/xml/en/procedures/20/20-012-014-tr.html>).

- Install the engine harness and the brackets. Refer to Procedures 019-043 (</qs3/pubsys2/xml/en/procedures/19/19-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System QSK19, QSK23, QSK45, QSK60 and QSK78 Engines, Bulletin 3666113 or 019-043 (</qs3/pubsys2/xml/en/procedures/13/13-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System Series QSK19 CM850 Modular Common Rail System Engine, Bulletin 4021493.
- Operate the engine and check for leaks.



with Electronically Actuated Injector

- Turn the keyswitch to the ON position for 30 seconds and allow the electric lift pump to prime the fuel system.
- Operate the engine and check for leaks.

